

# МЕТОД НЕИЗВЕСТНЫХ КОЭФФИЦИЕНТОВ

• ПРИМЕР:  $\frac{5x-6}{x-1} = \frac{A}{x-1} + B = \frac{A-B+Bx}{x-1}$

$$\begin{cases} B=5 \\ A=-1 \end{cases} \Rightarrow \frac{5x-6}{x-1} = \frac{-1}{x-1} + 5$$

• ПРИМЕР:  $x^2 - 6x - 3 = a(x-b)^2 + c = ax^2 - 2abx + ab^2 + c$

$$\begin{cases} a=1 \\ b=3 \\ c=-12 \end{cases} \Rightarrow x^2 - 6x - 3 = (x-3)^2 - 12$$

• ПРИМЕР:  $3x^3 - 2x^2 - x + 7 = Ax \cdot (Bx - C(Dx^2 + W)) + W = -ACD$

$$\begin{cases} W=7 \\ A \cdot C = -\frac{1}{7} \\ A \cdot B = -2 \\ A \cdot C \cdot D = -3 \end{cases} \Rightarrow \begin{cases} W=7 \\ A = \frac{1}{7} \\ C = -1 \\ B = -14 \\ D = 21 \end{cases} \Rightarrow 3x^3 - 2x^2 - x + 7 = \frac{x}{7} \cdot (-14)$$

• ПРИМЕР:  $\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

$$\sum_{i=0}^n i^2 = an^3 + bn^2 + cn + d \quad (\text{какой-то } p)$$

Заметим:  $p(1)=1 \Rightarrow a+b+c+d=1$

$$\sum_{i=0}^n i^2 - \sum_{i=0}^{n-1} i^2 = n^2 + \sum_{i=0}^{n-1} - \sum_{i=0}^{n-1} = n^2 \Rightarrow p(n) - p(n-1) = n^2;$$

$$n^2 = \cancel{an^3} + \cancel{bn^2} + \cancel{cn} + \cancel{d} - \cancel{an^3} + 3an^2 - 3an + a - \cancel{bn^2} + 2bn - b$$

$$n^2 = 3an^2 + n(2b-3a) + a+c-b$$

$$\begin{cases} a = \frac{1}{3} \\ b = \frac{1}{2} \\ c = \frac{1}{6} \\ d = 0 \end{cases} \Rightarrow$$

$$\sum_{i=0}^n i^2 = \frac{1}{3} \cdot n^3 + \frac{1}{2} n^2 + \frac{1}{6} n = \frac{2n^3 + 3n^2 + n}{6}$$